Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahaj-bhadja sahaj-bhadja

Summary

Backend engineer with 2+ years building cloud APIs, distributed services, and AI-backed platforms. Strong in Python, Java, SQL, AWS, CI/CD, and secure production systems.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Advanced NLP, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Architected **multi-tenant** AI education backend on **Azure SQL**, **Blob Storage**, and **FastAPI**, implementing row-level isolation, normalized schemas, and indexed queries for 500+ users, cutting dashboard latency from 1.2s to 350ms.
- Built **ETL-style** integration pipelines across **FastAPI**, **SQL Server**, **PostgreSQL**, and cloud storage, processing 5K+ monthly jobs with validation, reconciliation, and **CI/CD**-backed releases that sustained 99%+ production reliability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Engineered **Flask** backend services and **REST APIs** for an AI grading platform serving 10K+ daily requests, integrating secure file ingestion, grading workflows, and model orchestration across instructor-facing production systems.
- Designed concurrent batch-processing workflows on **AWS RDS** using multithreading and fault-tolerant execution, processing 1K+ files per batch while maintaining stable API behavior and reducing grading turnaround time by 60%.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered **microservice**-based data-sharing software with **Jenkins**, **PyTest**, and **Mockito** automation across Windows, Linux, and macOS, improving release reliability through 20+ unit and integration test suites in production.
- Optimized large-scale **API** and **ETL** workflows with **AWS Glue**, **S3**, and object-oriented design patterns, improving response times by 30% and increasing pipeline efficiency by 68% under production constraints.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Developed **MERN** platform features with **REST APIs**, **WebSockets**, and microservices, supporting 2M+ synchronized records across core business workflows while improving modularity, real-time updates, and maintainability during rollout.
- Automated Docker-based build and deployment workflows with **Jenkins CI/CD**, cutting release time by 50% and improving consistency, troubleshooting speed, and team-wide development practices across shared environments daily.

Skills

Languages: Python, Java, SQL, JavaScript, TypeScript, C++, Bash
Backend & APIs: FastAPI, Flask, Node.js, REST APIs, microservices, multithreading, background workers, distributed systems
Security & Cloud: authentication, authorization, policy enforcement, row-level security, identity infrastructure, audit logging, AWS, Azure, GCP, Docker, Kubernetes, AWS RDS, S3, Blob Storage
Data & Engineering: PostgreSQL, SQL Server, MySQL, MongoDB, ETL pipelines, query optimization, schema design, Jenkins, GitHub Actions, CI/CD, unit testing, integration testing, Git, code reviews, Agile/Scrum

Projects

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP*

- Engineered a secure document platform using **LLaMA-2**, **FAISS**, **Pinecone**, and **PostgreSQL**, enabling semantic retrieval across 50K+ records with sub-second search, summarization, and access-controlled collaborative workflows for teams.

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB, Plaid*

- Built a fintech backend supporting 50,000+ users and 1M+ transactions, combining **Kubernetes**, **MongoDB**, **Plaid**, secure token workflows, and real-time synchronization to maintain reliable high-volume transactional APIs daily.

Blockchain Authentication Analysis | *Blockchain Authentication, Access Control, Identity Assurance*

- Designed a **blockchain-based authentication** model for decentralized login and role-based access control, reducing repeated key verification steps by 50% while strengthening tamper-evident identity assurance across user-access scenarios.